

Catalytic Pyrolysis of Pinyon–Juniper Using Red Mud and HZSM-5

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S Supporting Information

ABSTRACT: Pinyon and juniper are invasive woody species in the western United States that occupy over 30 million hectares of land. The U.S. Bureau of Land Management (BLM) has embarked on harvesting these woody species to make room for range grasses for grazing. The major application of harvested pinyon–juniper (PJ) is low-value firewood. Thus, there is a need to develop new high value products from this woody biomass to reduce the cost of harvesting. We investigated the fractional catalytic pyrolysis of PJ using both HZSM-5 catalyst and red mud at 475 °C in a fluidized bed reactor at atmospheric pressure. Both the HZSM-5 and the red mud were effective catalysts for producing low-viscosity pyrolysis oils. Oils that were catalyzed with red mud had a lower viscosity (96 cP @40 °C) than oils that were catalyzed with HZSM-5 (213 cP @40 °C). In both cases, the yields of liquids ranged from 42 wt % to 49 wt %. The mechanisms of catalysis by the two catalysts were quite different. The HZSM-5 rejected oxygen mostly as carbon monoxide (CO) and produced lower amounts of carbon dioxide (CO₂); in contrast, the red mud produced more CO₂ and less CO. However, both catalysts produced similar amounts of water. The char/coke yields from both catalysts were similar but the total gas yields were slightly different. The higher heating value of the red mud catalyzed oil (HHV = 29.46 MJ/kg) was slightly higher than that catalyzed by HZSM-5 (HHV = 28.55 MJ/kg). Thus, red mud can be used to achieve similar catalytic pyrolysis results as HZSM-5 catalysts.

■ INTRODUCTION

Pinyon–juniper (PJ) woodlands occupy ~30 million hectares in the western United States, which includes the states of Arizona, California, Colorado, Idaho, Nevada, New Mexico, Oregon, Utah, and Wyoming.^{1,2} The tree species associated with PJ woodlands are extremely variable, with complex distribution patterns.³ The juniper trees found in the PJ woodlands consists of alligator juniper (*Juniperus deppeana* Steud.), one-seed juniper (*Juniperus monosperma* (Engelm.) Sarg.), western juniper (*Juniperus occidentalis* Hook.), Utah juniper (*Juniperus osteosperma* (Torr.) Little), and Rocky Mountain juniper (*Juniperus scopulorum* Sarg.). The pinyon trees consists of Mexican pinyon (*Pinus cembroides* Zucc.), pinyon (*Pinus edulis* Engelm.), and singleleaf pinyon (*Pinus monophylla* Torr. and Frem.).³

PJ woodlands domination decreases the herbaceous vegetation and increases bare lands which in turn increases soil erosion and nutrition loss.^{4,5} Studies have shown that expansion of PJ woodlands have reduced the magnitude of precipitation,⁶ and increased soil erosion by four times.⁷ PJ woodlands increase the potential of crown fires, which promote the infiltration of exotic species.⁸ Thus, land management agencies are focusing on reducing the population of PJ woodlands by bulldozing, chaining,⁹ hand cutting, mechanical removal,¹⁰ and prescribed fire. The harvested material is, therefore, a mixture of several species bundled together, which are used for low-value applications.

Catalytic pyrolysis is an emerging technology that has the potential of producing liquid fuels from biomass feedstocks. The major advantages of the catalytic pyrolysis process are improvement in the stability of the pyrolysis oils, because of cracking of the reactive species into either low-molecular-weight organics or gasification of some of this reactive species to carbon oxides and C1–C5 hydrocarbons, reduction in

viscosity, improved deoxygenation of the oils, and higher carbon content, which leads to higher heating values. The most effective catalysts used for biomass conversion to liquids are mostly acidic zeolites and modification of these catalysts. Although they produce relatively stable pyrolysis oils, they also generate considerable quantities of water, carbon oxides, and coke.^{11–13} This combination of gaseous products, coke, and water constitute a loss of hydrogen and carbon and a reduction in the overall carbon and hydrogen efficiencies. Thus, there is a pressing need to develop new catalysts that improve carbon efficiency and yet produce stable pyrolysis oils.

Red mud or red sludge is the solid residue from the processing of bauxite to alumina using the Bayer process. Red mud is a complex mixture of metallic oxides such as ferric oxide, aluminum oxide, titanium dioxide, magnesium oxide, calcium oxide, silicon oxide, and other minor compounds, in addition to the residual sodium hydroxide used in the extraction process. Catalytic applications of red mud have been extensively reviewed by Sushil and Batra.¹⁴ This material has been investigated by several researchers for catalytic applications such as hydrodechlorination,¹⁵ hydrogenation of anthracene,^{16,17} pyrolysis of waste plastics and waste oils,^{18,19} hydrogenation and liquefaction of rye straw,²⁰ upgrading of hemp-seed pyrolysis oils,²¹ and acetic acid.²² In most of these applications, red mud was activated through acid treatment and other modifications. However, there has not been any systematic investigation of using red mud as fractional catalytic pyrolysis catalysts. In this paper, we report the investigation of red mud as a catalyst for conversion of pinyon–juniper into pyrolysis oils for higher-value applications.

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EXPERIMENTAL METHODS

Materials. Pinyon–juniper (PJ) biomass chips were supplied by the U.S. Bureau of Land Management. The pinyon pine and juniper trees are usually harvested together; therefore, in order to represent the mixture, pinyon pine and juniper wood were manually mixed in a ratio of 50:50 (wt %). Wood samples were dried to equilibrium moisture content under ambient laboratory conditions and then ground in a Wiley mill (model 4) until all the biomass passed through a 1-mm screen.

A commercial catalyst, ZSM-5, was obtained from BASF Catalysts LLC (Iselin, NJ, USA) and calcined at 550 °C for 5 h, to convert ZSM-5 into protonated form (HZSM-5). The catalyst was sieved to obtain particle size of 125–180 μm . The Brunauer–Emmett–Teller (BET) method was used to measure the surface areas of the catalyst using Quantachrome BET surface analyzer (Quantachrome Instruments, Boyton Beach, FL).

The red mud supplied by Sherwin Alumina Co LLC, (Gregory, TX), was dried at ambient laboratory conditions, ground with mortar and pestle and sieved to appropriate particle size (125–180 μm) for fluidized bed pyrolysis studies. The composition of the red mud was determined by X-ray diffraction (XRD) using X'Pert Pro XRD spectrometer (PANalytic Inc., Westborough, MA) and X-ray fluorescence (XRF) using Phillips PW2404 XRF spectrometer (PANalytic Inc., Westborough, MA) and its surface area was determined using the Quantachrome BET surface analyzer.

Biomass characterization. The biomass was characterized on the basis of moisture, ash content, elemental composition, and higher heating value (HHV) (see Table 1). The moisture content of the

Table 1. Composition and Higher Heating Value of Pinyon–Juniper Biomass

parameter	value
composition (wt %)	
moisture	7.55 $\pm$ 0.06
ash content	0.46 $\pm$ 0.04
carbon	53.43 $\pm$ 0.11
hydrogen	6.58 $\pm$ 0.04
nitrogen	0.16 $\pm$ 0.01
sulfur	0.00
oxygen ^a	39.37 $\pm$ 0.09
experimental higher heating value, HHV (MJ/kg)	19.54 $\pm$ 0.16

^aOxygen by difference.

biomass was determined according to standard method ASTM E1756-08;²³ the ash content of the biomass was determined using ASTM E1755-01 standard method;²⁴ the elemental composition (C, H, N, S, and O) of biomass was determined using the organic elemental analyzer (Flash 2000, ThermoFischer Scientific CHNS-O analyzer), but the oxygen was calculated by difference. The higher heating value (HHV) of the samples was determined using a IKA Model C2000 basic bomb calorimeter (IKA Works, Inc., Wilmington, NC, USA).

Pyrolysis of Biomass. The pyrolysis of the biomass feedstocks was conducted in a 51-mm (2-in.) fluidized bed reactor described in detail by Agblevor et al.¹² and Mante et al.²⁵ Thus, the ground biomass was fed into a hopper and conveyed by K-Tron screw feeder into an entrainment zone, where it was entrained by nitrogen carrier gas and transported through an air-cooled feeder tube into the bubbling fluid bed. The bubbling fluid bed medium was either sand or catalyst (100 g), depending on the desired outcome. The pyrolysis was conducted at 475 °C at atmospheric pressure. The fluidizing gas was nitrogen at a flow rate of three times the minimum fluidization velocity for catalytic (6.5 L/min) and noncatalytic pyrolysis (15 L/min). The biomass feed rate was 150 g/h, and weight hourly space velocity was 1.5 h⁻¹. The pyrolysis products exiting the reactor were first separated with a hot gas filter to collect the solid residue and some entrained catalyst. The vapors and noncondensable gases exiting the hot gas filter were then passed through two ethylene-glycol-cooled condensers that were

maintained at 10 and 4 °C, respectively, to condense the pyrolysis vapors. The noncondensable gases and aerosols were then passed through an electrostatic precipitator (ESP) maintained at 20 kV to condense the aerosols. The clean noncondensable gases (NCG) were then passed through a coalescing filter and a totalizer and sampled online with a Varian MicroGC gas chromatographic analyzer (Agilent Technologies, Santa Clara, CA, USA). The excess gases were vented through a flare. All the reaction apparatus (reactor, hot gas filter, condensers, electrostatic precipitator) were weighed before and after each experiment. This enabled gravimetric determination of the yield of various products. The pyrolysis times were 120–180 min. In order to determine the catalyst activity, oil samples were collected at 60-min intervals and their viscosities were measured. In addition, the gaseous products were analyzed every 7 min to follow the composition of the NCG, which is related to the catalyst activity. All experiments were conducted in triplicate to determine reproducibility. In addition, some red mud catalyst was regenerated after the first run and reused for the pyrolysis.

Gas Analysis. A fraction of the noncondensable gases (NCG) exiting the totalizer (0.1 L/min) was connected to a Varian 490-microGC system for online analysis. The NCG was automatically sampled every 7 min for the duration of the experiment. The microGC was equipped with two modules: a 10 m Molsieve 5A (MS) column, and a 10 m porous polymer (PPU) column. Each module had a thermal conductivity detector. MS was used to analyze hydrogen, methane, and carbon monoxide (CO), while carbon dioxide (CO₂) and C₁–C₅ hydrocarbons were analyzed on the PPU column.

Bio-oil Analysis. The physicochemical properties of the pyrolysis oil samples were characterized using several methods described below. The pH was measured using Mettler Toledo pH Meter and probe (Mettler–Toledo GmbH, Switzerland). The pH data were obtained after 5–10 min stabilization of the mechanically stirred oil. The dynamic viscosities and densities of the oils were measured at 40 °C using Anton Parr Stabinger viscometer svn 3000 (Anton Paar USA, Inc., Ashland, VA, USA). A Metrohm 701KF Titrino (Brinkmann Instruments, Inc., New York, USA) and 703 titration stand setup were used for the volumetric Karl Fischer titration. Hydranal-Composite 5 reagent was used for the titration. Fifty milliliters (50 mL) of methanol was placed in the titration vessel and conditioned. Approximately 60–100 mg of the oil sample was loaded into a hypodermic plastic syringe and weighed. The sample was injected into the titration solvent and the syringe was reweighed. The water content was titrated volumetrically and the percentage mass recorded. The elemental composition (CHNOS) was determined using ThermoFischer Scientific Flash 2000 organic elemental analyzer, but the oxygen content was calculated by difference; the higher heating value (HHV) was determined using a IKA Model C2000 basic bomb calorimeter. The ¹³C NMR spectra were recorded on a JEOL 300 MHz NMR spectrometer (JEOL, Ltd., Tokyo, Japan). About 1.0 g of oil was dissolved in 0.6 mL deuterated dimethyl sulfoxide-d₆ (DMSO-d₆) in a 5-mm sample probe. The DMSO-d₆ containing 1% (v/v) tetramethylsilane (TMS) was obtained from Sigma–Aldrich (Sigma–Aldrich, St. Louis, MO, USA). The observing frequency for the ¹³C nucleus was 100.58 MHz, the pulse width was 10 μs , the acquisition time was 1.58 s, and the relaxation delay was 2 s. The spectra were obtained with 3000 scans and a sweep width of 20 kHz.

RESULTS AND DISCUSSION

Catalyst Characterization. The BET surface area of the HZSM-5 catalyst used for the pyrolysis was 115 m²/g. The red mud water slurry had a pH 9 and the dry ground material used for the experiments had a BET surface area 30 m²/g. The XRF analysis showed that the red mud was composed of the following major metal oxides (wt %): Fe₂O₃ (53.98), Al₂O₃ (13.53), SiO₂ (8.91), CaO (8.87), TiO₂ (6.18), and Na₂O (5.83). The presence of these oxides was further confirmed with XRD analysis. There was no activation of the red mud before being used, thus, the reported effect is on an as-received

basis. However, the red mud samples were heated in the reactor for ~ 2 h to get to a stable pyrolysis temperature before each reaction. The red mud did not have any magnetic properties before the pyrolysis, but after pyrolysis the residue had magnetic properties. The change in properties of the red mud was attributed to the conversion of hematite (Fe_2O_3) to magnetite (Fe_3O_4) under the pyrolysis conditions (Figure 1).

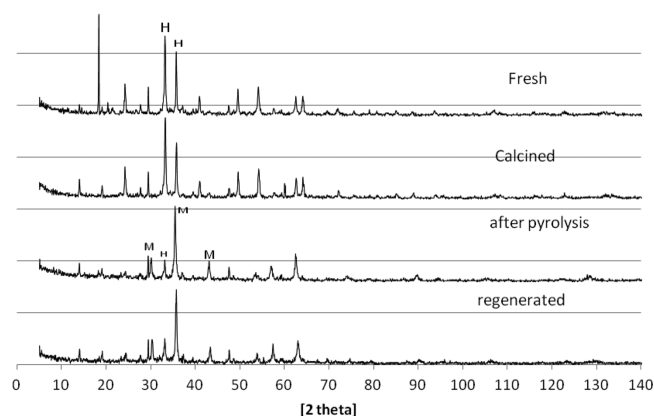


Figure 1. Comparison of XRD pattern of different red mud samples. [Legend: H = hematite, M = magnetite.]

The 2θ signals before and after the pyrolysis were typical of hematite and magnetite.²⁶ It has been shown by Jozwiak et al.²⁷ that, under 5% hydrogen concentration or 5% CO concentration, hematite is reduced to magnetite between 150 °C and 400 °C. The CO concentration in the total PJ pyrolysis gas (NCG plus nitrogen) was 2.7 vol %, and the hydrogen was <0.1 vol %; these gas concentrations were not optimal for hematite reduction, but they were sufficient for partial conversion of hematite to magnetite which resulted in a mixture of oxides, as shown in Figure 1. The role of each of these iron oxides in the catalysis of the biomass is not well understood. When a similar pyrolysis reaction was conducted using pure Fe_2O_3 (data not reported), the reduction in viscosity of the pyrolysis oil was much less, which suggested that other components in the red mud were also participating in the catalysis of the biomass during pyrolysis.

In order to study the effect of preheating the red mud in the reactor before the pyrolysis, the red mud was calcined at 550 °C for 5 h in a furnace; the XRD pattern obtained (Figure 1) shows that no magnetite was formed. The calcination, pyrolysis, and regeneration introduced other changes in the structure of the red mud, as shown in Figure 1. The 2θ signals between 10° and 20° had reduced intensities after calcination, regeneration, or pyrolysis, but the compounds responsible for the changes were not identified. Furthermore, the roles of these structural changes in the catalytic activity of the red mud are not understood. It is also interesting to note that the calcinations of the red mud did not influence the hematite properties of the

iron oxide, because the atmosphere was oxidative (Figure 1). Similarly, the catalyst regeneration during which the coke/char was burned off the surface of the catalysts was also oxidative and did not cause any phase change of the catalyst; thus, the regenerated catalyst had composition similar to that of the red mud after pyrolysis (Figure 1).

Pyrolysis Products Distribution. The pyrolysis products distribution for PJ biomass using various pyrolysis media is shown in Table 2 for pyrolysis lasting 2 h. The liquid products were segregated in the three condensers: most of the aqueous fraction was found in the first condenser, while the oil collected in the electrostatic precipitator (ESP) had low moisture content. However, the properties of the organic phase of the condenser liquids were similar to those from the ESP oils. In the case of the silica sand (sand) pyrolysis, the aqueous phase was miscible with the organic fraction whereas in the cases of HZSM-5 and red mud the aqueous phases were immiscible with the organic phases because the oils were more deoxygenated, which made them less polar than the sand pyrolysis oil (see Table 4, presented later in this work).

The HZSM-5 and red mud pyrolysis decreased the total liquid yield, relative to that from the sand. The loss in total liquid yield was mostly due to the production of large amounts of gaseous products (carbon oxides) from the organic fraction. The reduction in the organic liquid yield was attributed to deoxygenation and cracking of the pyrolysis vapors by the catalysts. The conversion of the organic liquid to water and char/coke by the catalysts was relatively low, compared to the amount of gas produced. The water yields of HZSM-5 and red mud were only slightly different from that due to the sand (see Table 2).

The char/coke yields for the HZSM-5 and the red mud were similar but slightly higher than that for the sand (Table 2). These char/coke yields were relatively low and followed trends reported by Agblevor et al.,^{11,12} because, in this process, the species in contact with the catalyst are unstable pyrolysis intermediates, whereas in traditional pyrolysis upgrading systems, the compounds in contact with the catalyst are stable molecules that promote coke formation. The pyrolysis medium greatly influenced the yield of gaseous products. The sand produced the least amount of gas, whereas the red mud produced the highest amount of gas, because of its ability to crack the pyrolysis vapors.

Properties of Catalytic Pyrolysis Oils. Because the ESP captured most of the oil product with minimal water content, this fraction was used for the characterization studies. The oil phase from the second condenser had similar properties, like those of the ESP oil; therefore, data from the ESP oils were representative of the total oils produced from this process. The physicochemical properties of the oils from the ESP are shown in Table 3. Clearly, it can be seen that the moisture content of the ESP oils were very low, because of the efficient condensation of the aqueous fractions in the ethylene-glycol-cooled condensers. The pH of red mud oil was higher than

Table 2. Pyrolysis Products Distribution of Pinyon–Juniper Biomass Pyrolyzed Using Various Pyrolysis Media

pyrolysis medium	product distribution (wt %, dry basis)				
	total liquid	organic liquid	water	char	gas
sand	60.31 $\pm$ 0.05	40.15 $\pm$ 0.51	20.16 $\pm$ 0.55	22.49 $\pm$ 0.39	17.21 $\pm$ 0.34
HZSM-5	49.34 $\pm$ 0.14	25.92 $\pm$ 0.11	23.42 $\pm$ 0.25	20.98 $\pm$ 0.32	29.69 $\pm$ 0.18
red mud	44.37 $\pm$ 1.02	20.85 $\pm$ 1.0	23.53 $\pm$ 0.02	23.50 $\pm$ 0.21	32.14 $\pm$ 1.23

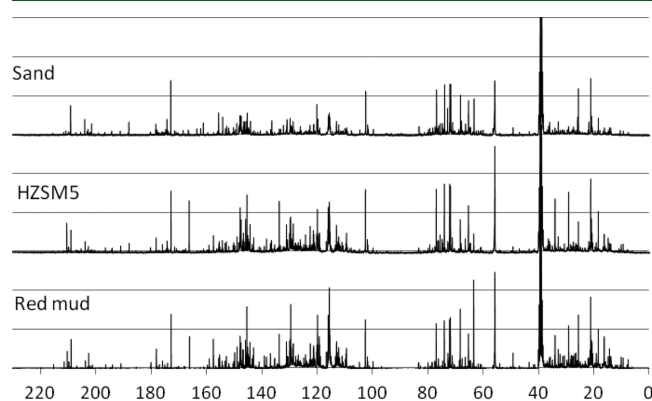
Table 3. Physicochemical Properties of Pyrolysis Oils Collected from the Electrostatic Precipitator

property	sand	HZSM-5	red mud
water content (wt %)	2.81 ± 0.15	1.72 ± 0.08	1.45 ± 0.08
pH	2.75	2.80 ± 0.10	3.56
density (g cm ⁻³)	1.22 ± 0.02	1.19 ± 0.04	1.14 ± 0.01
dynamic viscosity (cP)	686.02 ± 27.52	213.68 ± 10.14	96.99 ± 19.54
carbon content (wt %)	58.11 ± 0.07	62.79 ± 0.10	65.13 ± 0.05
hydrogen content (wt %)	6.72 ± 0.04	6.86 ± 0.04	7.17 ± 0.01
nitrogen content (wt %)	0.13 ± 0.02	0.27 ± 0.03	0.21 ± 0.02
oxygen content (wt %)*	35.04	30.08	27.49
higher heating value, HHV (MJ/kg)	24.87 ± 0.37	28.55 ± 0.23	29.46 ± 0.62

*Oxygen by difference.

those of the sand and HZSM-5 oils, probably because of the improved decarboxylation of some of the acidic groups. The densities of the red mud oils were also slightly lower than those of the sand and HZSM-5, although they all had densities greater than that of water. The dynamic viscosities of the oils showed dramatic differences. The red mud oils were ~7 times less viscous than the sand oils, whereas the HZSM-5 oils were only 3 times less viscous. These major differences were attributed to the cracking of the oils by the red mud and the HZSM-5, which resulted in high gas production. The relative carbon content of the red mud oils was 14% higher than that of the sand oil, and its oxygen content was 25% lower than that of the sand oil. The hydrogen content of the red mud pyrolysis oils were also slightly higher than those of the sand oil and HZSM-5 oils, probably because most of the oxygen was rejected as carbon dioxide and less as water. There were significant differences in the higher heating values (HHV) of the oils. Both the red mud and HZSM-5 oils had much higher HHV than the sand oil.

The functional groups present in the pyrolysis oils were characterized by semiquantitative integration of the ¹³C NMR spectra. The ¹³C NMR spectra of sand, HZSM-5, and red mud are shown in Figure 2. The chemical shifts in the spectrum were assigned to different functional groups according to the analysis reported in Mante et al.²⁸ The semiquantitative analysis of the

**Figure 2.** ¹³C NMR spectra of bio-oil from catalytic and noncatalytic pyrolysis oils.

¹³C NMR functional groups show that pyrolysis with sand produced more highly oxygenated oils that had higher amounts of carbohydrate degradation products, alcohol, ethers, methoxylated phenols, carboxylic groups, aldehydes, and ketones (Table 4). The catalytic pyrolysis oils had lower

Table 4. Functional Group Distribution of Catalytic and Noncatalytic Pyrolysis Oils from ¹³C NMR Spectral Integration

functional groups	chemical shift, δ (ppm)	sand	HZSM-5	red mud
aliphatics	0–55	17.70	20.21	26.03
methoxyl in lignin	55–57	8.22	6.72	5.04
levoglucosan	60–105	30.04	16.08	12.45
aromatics	105–160	38.03	53.74	53.93
carboxylic acids	160–180	3.73	1.99	1.51
ketones and aldehydes	180–210	2.29	1.26	1.05

concentrations of carbohydrate degradation products, alcohols and ethers, while the aliphatics and aromatic products were relatively higher than that from sand. The carboxylic acid, ketone, and aldehyde concentrations were also relatively lower than that of the sand.

The comparison between HZSM-5 and red mud pyrolysis oils spectra showed that the latter was more effective in the conversion of the carbohydrate degradation products into other compounds (Table 4). The relative amounts of aromatic compounds produced by the red mud were similar to HZSM-5, but there were more aliphatic carbon signals in the red mud oils than the HZSM-5 oils. The increased aliphatic signal intensity was attributed to increased demethoxylation of the lignin moieties, which resulted in the production of methanol and subsequent alkylation reactions. The methanol signal at 50 ppm was much higher for the red mud oil than the other two oils (Figure 2). It has been reported that phenol can be alkylated with methanol to produce cresols and anisole in the presence of zeolite and other acidic catalysts^{29,30} some of these compounds have been reported in fractional catalytic pyrolysis studies by Agblevor et al.^{11,12} Para-cresol has also been obtained from vapor-phase alkylation of phenol with methanol on MgO-based catalyst at 500 °C.³¹

Pyrolysis Gases. The pyrolysis gas components were similar for all the pyrolysis media, but varied in their individual concentrations. All pyrolysis gases contained CO, CO₂, and C₁–C₅ hydrocarbons (Table 5) with CO and CO₂ constituting 84%–90% of all gases. The CO₂ contents were highest for the

Table 5. Yield^a of Various Gases Using Different Pyrolysis Medium

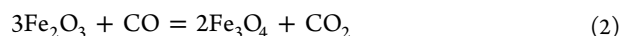
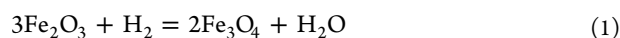
gas	sand	HZSM-5	red mud
hydrogen	0.14 ± 0.02	0.19 ± 0.02	0.49 ± 0.12
methane	0.50 ± 0.05	1.21 ± 0.08	1.59 ± 0.16
carbon monoxide	5.41 ± 0.81	14.27 ± 0.56	9.21 ± 0.27
carbon dioxide	10.30 ± 0.34	10.85 ± 0.05	18.51 ± 0.41
ethane	0.09 ± 0.01	0.67 ± 0.02	0.36 ± 0.01
propane	0.31 ± 0.02	0.64 ± 0.01	0.37 ± 0.04
<i>n</i> -butane	0.25 ± 0.14	2.06 ± 0.05	1.37 ± 0.24
<i>n</i> -pentane	0.21 ± 0.02	0.27 ± 0.06	0.23 ± 0.03
total	17.21 ± 1.37	29.69 ± 1.01	32.14 ± 1.28

^aThe yield is expressed as a percentage of the biomass feed (wt %, dry basis).

sand and red mud pyrolysis gases. In contrast HZSM-5 pyrolysis gases had the highest CO content and red mud pyrolysis gases had the lowest CO content. The main difference in the carbon oxide contents of the gases may be due to differences in reaction pathways. In the case of the sand, the carbon oxides production was attributed to thermal cracking, whereas in the cases of HZSM-5 and red mud, there appeared to be catalytic cracking. The catalytic rejection of oxygen from the biomass appeared to be through production of H₂O, CO, and CO₂. The catalytic CO and CO₂ production from biomass pyrolysis could be through decarbonylation, ketonization, decarboxylation, or all three pathways. In the case of the HZSM-5, it appeared the decarbonylation reaction was dominant, producing more CO than CO₂. Our results are in agreement with Stefanidis et al.,³² who reported that acidic catalysts favored decarbonylation reactions to produce CO.

In the case of the red mud catalysis, the CO₂ production was more pronounced than that of CO. The CO₂ production from the red mud was attributed to catalytic activity instead of thermal cracking, because the organic liquids produced had lower oxygen, higher carbon, and lower viscosity than those produced from the sand (recall Table 3). Oxygen rejection from the pyrolysis products was probably through decarboxylation reactions, because the relative amounts of carboxylic acids and ketones were lower for the red mud oil, compared to the HZSM-5 and sand oils (Table 4). The ¹³C NMR carboxylic acid signal for the red mud oil was less than one-half that from the sand oil. Oxygen rejection through CO₂ has been reported as through ketonization reactions, which converted carboxylic acids to ketones and released CO₂.^{33–35} Thus, if the high CO₂ production from the red mud reaction proceeded through a ketonization pathway, then we should have observed an increase in the ¹³C NMR ketone signals and a decrease in the carboxylic carbon signal relative to the sand, but this was not observed.

The higher CO₂ production from the red-mud-catalyzed pyrolysis could also be due to a contribution from the reduction of hematite to magnetite. Jozwiak et al.²⁷ reported that under low hydrogen and CO concentrations at 150–400 °C and atmospheric pressure, hematite was reduced to magnetite with the subsequent release of CO₂ and H₂O, as shown in eqs 1 and 2:



The pyrolysis gas composition showed the presence of both H₂ and CO and the red mud after pyrolysis had magnetic properties; thus, it is plausible that some of the CO₂ originated from the reduction of hematite to magnetite. The XRD data of the red mud after the pyrolysis also confirmed the formation of magnetite (see Figure 1).

The contribution of CO₂ from the reduction of hematite to magnetite was presumed minimal because reaction 2 is very rapid and the CO₂ will be released within a few minutes from the start of the pyrolysis reaction. Furthermore, this assertion is supported by the regenerated catalyst (magnetite/hematite) pyrolysis data, which had similar CO/CO₂ ratios for the fresh and regenerated catalyst pyrolysis (Figure 3). During the red mud catalytic pyrolysis, the catalyst was partially transformed into magnetite/hematite and regeneration did not cause any phase change in the catalyst because of the oxidative nature of the process (Figure 1).

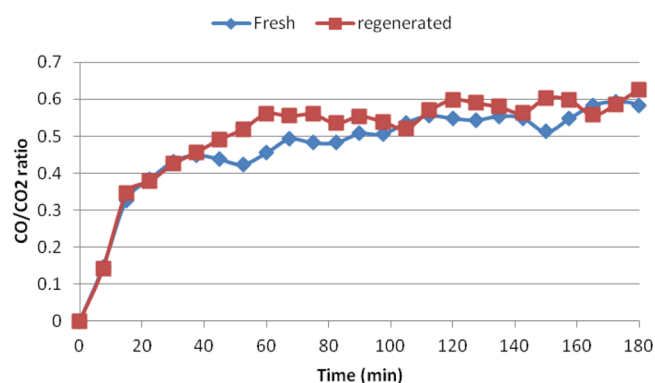


Figure 3. CO/CO₂ ratio from PJ pyrolysis using fresh and regenerated red mud.

Oxygen rejection as H₂O was similar for the two catalysts since the H₂O yields were similar and higher than that of the sand. The hydrogen yield also varied, according to the pyrolysis medium. The highest hydrogen production was from the red mud medium, whereas the sand produced the lowest amount of hydrogen. The concentrations of hydrocarbons also increased for catalytic pyrolysis compared to noncatalytic pyrolysis. Butane accounted for ~39% of hydrocarbons for both catalysts and methane accounted for 27% and 37% of hydrocarbons, respectively, in HZSM-5 and red mud.

Catalyst Deactivation and Regeneration. The activity of the catalyst was monitored through the variation of CO and CO₂ concentration over time.^{11,12} The CO/CO₂ ratios variation (wt% basis) for sand, HZSM-5, and red mud during the course of pyrolysis are shown in Figure 4. The average CO/

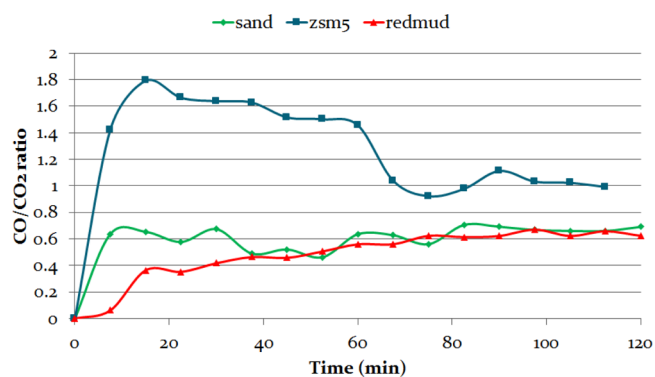


Figure 4. CO/CO₂ ratio (wt %) for sand, HZSM-5, and red mud during the time course pyrolysis of PJ biomass.

CO₂ ratio for sand was constant at ~0.6 throughout the pyrolysis process. The ratio of CO/CO₂ for HZSM-5 was high during the first hour (1.8–1.4) but decreased during the second hour to a constant value of 1.0. Thus, the catalyst was most active during the first 60 min, but the activity decreased sharply during the second hour and remained constant thereafter. Since the CO/CO₂ ratio did not decrease to the value of the sand, this suggested that there was some residual activity in the catalyst. During the first 60 min of pyrolysis, no significant amount of oil accumulated in the ESP and most of the product was gas, but after 120 min a significant amount of oil accumulated, which suggested that most of the oil was produced during the partially deactivated phase of the catalyst.

Table 6. Physicochemical Properties of Pyrolysis Oils from Fresh and Regenerated Red Mud

property	Fresh Red Mud			Regenerated Red Mud		
	60 min	120 min	180 min	60 min	120 min	180 min
water content (wt %)	0.67 ± 0.11	0.93 ± 0.08	0.99 ± 0.07	1.80 ± 0.08	1.42 ± 0.02	1.15 ± 0.02
pH	3.41 ± 0.04	3.58 ± 0.06	3.49 ± 0.10	3.76 ± 0.05	3.65 ± 0.06	3.58 ± 0.05
density (g cm ⁻³)	n/d	1.26 ± 0.11	1.19 ± 0.10	n/d	1.14 ± 0.15	1.14 ± 0.18
dynamic viscosity (cP)	n/d	101.16 ± 8.79	120.25 ± 6.91	n/d	99.34 ± 6.44	122.81 ± 9.7
carbon content (wt %)	67.63 ± 0.41	67.10 ± 0.21	67.16 ± 0.09	66.03 ± 0.23	66.47 ± 0.08	66.55 ± 0.02
hydrogen content (wt %)	7.02 ± 0.12	7.09 ± 0.05	7.13 ± 0.02	7.21 ± 0.04	7.23 ± 0.01	7.27 ± 0.03
nitrogen content (wt %)	0.31 ± 0.03	0.29 ± 0.05	0.28 ± 0.03	0.24 ± 0.01	0.25 ± 0.02	0.23 ± 0.01
oxygen content (wt %) ^a	25.04	25.52	25.43	26.55	26.05	25.95

^aOxygen by difference.

The final oil collected after 120 min had a viscosity that was ~3-fold less than that of the sand pyrolysis oil (Table 3).

The red mud produced more CO₂ than CO and, thus, the CO/CO₂ ratio was below 1.0 throughout the experiment. During the first 60 min, CO/CO₂ ratio was lower than that of the sand, but the ratio increased gradually and was only slightly less than the sand ratio after 40 min. Even though the CO/CO₂ ratios of the sand and red mud were similar after 60 min of pyrolysis, the properties of the composite pyrolysis oils (total oils collected between 0 and 120 min) were significantly different (Table 3). The red mud pyrolysis oil was ~7 times less viscous than the sand pyrolysis oil, which suggested that the reaction pathway was different from that of the sand.

After 3 h of continuous pyrolysis using red mud, the CO/CO₂ ratio and the viscosity of the oil suggested that the catalyst had been deactivated significantly. The catalyst was then regenerated by burning the char/coke deposit in a muffle furnace at 550 °C for 5 h. The regenerated catalyst had magnetic properties, which suggested that the hematite (Fe₂O₃) had been reduced to magnetite (Fe₃O₄) under the pyrolysis conditions. To verify that the change in properties of the red mud was not due to the regeneration process, which is oxidative, cooled freshly discharged catalyst from the pyrolysis reactor was tested for magnetic activity; this showed positive magnetic activity. In addition, fresh red mud was calcinated at 550 °C for 5 h and tested for magnetic activity, but this material did not possess any magnetic activity. The formation of magnetite was further confirmed by comparing the XRD spectra of fresh, regenerated, and calcinated red mud (Figure 1). Both hematite and magnetite have 2θ signals at 35.2°, but for the regenerated catalyst, the intensity of this signal increased significantly. Furthermore, only magnetite has signal at 2θ = 30° and hematite has a 2θ signal at 33°. The signals for both hematite and magnetite were detected in the regenerated catalyst confirming that some of the hematite was converted to magnetite under the pyrolysis conditions.

The pyrolysis product distributions of fresh and regenerated red mud were similar (data not shown). The physicochemical properties of ESP oil from fresh and regenerated red mud collected at 60 min intervals during the pyrolysis are shown in Table 6. There was not much difference in pH, densities, elemental composition, and energy content of the oils collected at different times for fresh red mud. The viscosity of the oil however increased with time, and this suggested that fresh red mud lost some of its activity to break down larger molecules.

The physicochemical properties of ESP oil from regenerated red mud showed that there was not much difference in density, pH, elemental composition, and energy content of the oils collected at different time intervals. As observed for the fresh

red mud pyrolysis oils, the viscosity of oil from regenerated red mud also increased with the time on stream. Thus, with fresh and regenerated red mud, there was not much difference in the physical properties of the oils.

The chemical compositions of the pyrolysis oils from fresh and regenerated red mud were analyzed with ¹³C NMR spectroscopy. Figures 5 and 6 show the ¹³C NMR spectra of

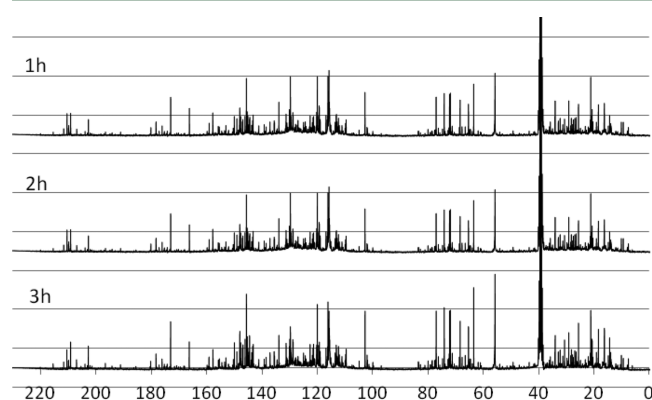


Figure 5. ¹³C NMR spectra of pyrolysis oils collected at different time intervals from pyrolysis of PJ biomass using fresh red mud.

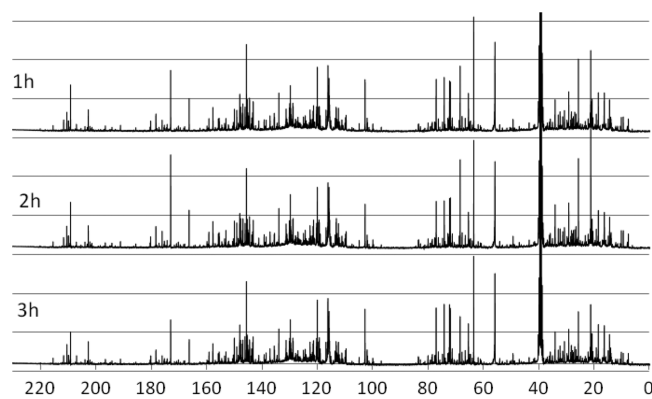


Figure 6. ¹³C NMR spectra of pyrolysis oils collected at different time intervals from pyrolysis of PJ biomass using regenerated red mud.

fresh and regenerated red mud collected at different time intervals. The semiquantitative integration of the ¹³C NMR spectra are shown in Table 7. Clearly, the two oils were similar in the relative concentration of most functional groups, thus suggesting that the catalyst activity was fully restored after the regeneration. Furthermore, the variation in the CO/CO₂ ratio (Figure 3) of the regenerated red mud and the fresh red mud

Table 7. Functional Groups Distribution (^{13}C -NMR Integration Data) of Pyrolysis Oils Collected at Different Time Intervals from the Pyrolysis of PJ Biomass Using Fresh and Regenerated Red Mud

chemical shift, δ (ppm)	Fresh Red Mud			Regenerated Red Mud		
	1 h	2 h	3 h	1 h	2 h	3 h
0–55	22.77	26.16	21.68	23.58	23.02	22.93
55–57	3.85	4.61	5.37	5.14	5.16	5.51
60–105	9.95	10.61	12.74	11.69	11.48	13.66
105–160	60.57	55.79	57.39	56.72	56.82	55.37
160–180	1.57	1.75	1.72	1.49	2.30	1.49
180–210	1.28	1.08	1.09	1.38	1.23	1.04

also support the assertion that the regenerated catalyst had similar activities like the fresh red mud. However, the regeneration was for only one cycle, so it is not clear how many regenerations can be accomplished before the red mud deactivates permanently.

DISCUSSION

The comparative catalytic pyrolysis studies of HZSM-5 and red mud clearly demonstrated that red mud can catalyze the fractional catalytic pyrolysis of biomass to generate pyrolysis oils with relatively lower viscosities than the HZSM-5 catalyst. The viscosities of the pyrolysis oils generated using the two catalysts were significantly different from each other and the baseline sand pyrolysis oils. The red mud pyrolysis oils were ~ 7 times less viscous than the baseline sand oils, while the HZSM-5 oils were only 3 times less viscous than the sand oils, even though the differences in their oxygen contents were not drastically different. In our recent studies, we showed that a major factor contributing to the increase in viscosity of pyrolysis oils during storage was the oligomeric products content.³⁷ The oligomeric products isolated from the oils derived from both carbohydrate and lignin decomposition products. These methanol-soluble oligomeric products were either solid or gel-like at room temperature. The higher the oligomeric products content, the higher the viscosity of the bio-oil. Applying this same concept to the HZSM-5, red mud, and sand oils in the current studies, this suggested that the red mud was more effective in decomposing the oligomeric products in the oil compared to the sand and HZSM-5. The other disadvantage of the HZSM-5 catalyst was the production of large quantities of gas during the early stages of the pyrolysis when the catalyst was extremely active and thus reducing liquid yields. The rejection of oxygen was also through the production of CO, which led to an excessive loss of carbon, thus reducing the overall carbon efficiency of the process.

Although the red mud pyrolysis process also generated large quantities of gas similar to the HZSM-5 pyrolysis, despite its lower surface area; the gas was richer in CO_2 than the HZSM-5 gas and the oils had slightly lower oxygen content than the HZSM-5 pyrolysis oils. The catalytic activity of the HZSM-5 appeared to be more dependent on the internal surface area, and once these were fouled with coke deposits, the reaction rate decreased considerably. In contrast, the red mud catalytic activity appeared to be dependent on the external surface area, which was ~ 3.8 times less than the HZSM-5 and yet the reactivity was maintained for ~ 3 h. Since there was constant attrition of the particles in the fluid bed, the fouling of the exterior surface of the red mud was reduced considerably,

compared to the HZSM-5; hence, better quality oil was produced using this catalyst. The viscosity of pyrolysis oils generated from the red mud catalysis process also increased during the pyrolysis time on stream; however, the increase was less rapid than the HZSM-5. Both catalysts appeared to reach a steady or equilibrium state during which the rejection of carbon oxides did not vary very much with the time on stream. The major advantage of the red mud catalyst is its apparent rejection of oxygen through CO_2 instead of CO and, hence, less consumption of carbon for oxygen rejection. The physico-chemical properties of the red mud oil also appeared to be slightly better than the HZSM-5 oils. Furthermore, red mud is a waste product of the Bayer alumina process and could therefore be procured at minimal cost and, hence, improve the economics of biomass catalytic pyrolysis process. Red mud could therefore be potentially used to replace HZSM-5 for catalytic biomass pyrolysis.

CONCLUSIONS

Pyrolysis of pinyon–juniper biomass was demonstrated using silica sand, HZSM-5, and red mud as effective pyrolysis media. The HZSM-5 and red mud had a catalytic effect on the pyrolysis of the feedstock, resulting in oils with much lower viscosity than the oils produced using silica sand as the pyrolysis medium. The pyrolysis oils generated using the red mud had viscosities that were ~ 7 times lower than the oils produced using the silica sand. The reaction pathway of the red mud was quite different from that of the HZSM-5 and silica sand, in that red mud rejected oxygen from the biomass pyrolysis vapors probably through decarboxylation reaction, instead of the decarbonylation observed for the HZSM-5. The red mud appeared to be stable and regenerable for catalytic pyrolysis reactions.

ASSOCIATED CONTENT

Supporting Information

This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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